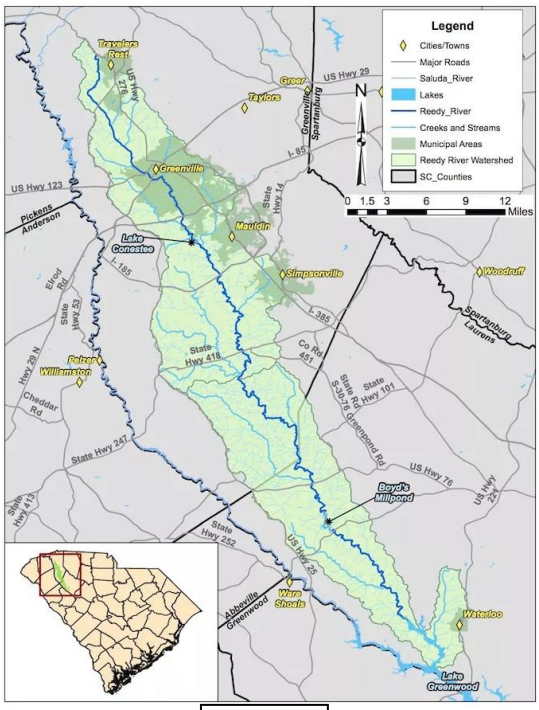
Hydrology of the Reedy River within Greenville County

Introduction

The Reedy River begins its journey just north of Travelers Rest, South Carolina within Greenville County. The river then continues its journey through the county and into the neighboring county of Laurens, where it connects with Lake Greenwood (Map 1). The total path of the river is about 65 miles long and has a drainage basin of 402 mi^2 [2]. The Reedy River is a part of the greater Saluda River Watershed, which is a HUC 8 watershed that spans from the boarder of South Carolina and North Carolina southeast to just outside of Columbia, South Carolina in the middle of the state. While the Reedy River might not be a long as the Saluda, its impact on South Carolina is clearly seen in the growth and development of the largest urban area in South Carolina, Greenville.

The Reedy River was originally used by Native American tribes like the Catawba and Cherokee tribes to hunt and fish [2]. As time moved on these tribes were replaced by the first colonial setters whom relied on the river to provide fish and water for plantations [2]. This use of the river for water transitioned quickly once industrialization occurred, which drove many mills to develop along with river's edge to power machines [2]. By the 1900's there were 8 mills operating on the river within two miles of the central city of Greenville [2]. The Reedy River slowly transitioned away from its ability to power machines, to its "most useful quality" of being able to wash away waste produced by these mills [2]. The ability for Greenville to discharge pollutants created by the textile mills, allowed Greenville to grow during this time period [2]. This pollution came to ahead when Conestee Mills, downstream of Greenville, sued the city of Greenville for the way the city was treating the river and damaged placed on their own mill, and its operations [2]. The case started in 1925, but was not ruled on by the South Carolina Supreme Court until 1931, "by then the city had built its first wastewater treatment plant in 1928" [2].



Map 1.

The damages to the Reedy River were understood back in 1929, when Furman University, established just south of the start of the river, opened its garden and arboretum in efforts to start the rehabilitation of the river [2]. Even by understanding the damages to the river, large projects like the construction of Interstate 85, an Army Air Base, and large residential developments led to massive amounts of sediment and silt to flow into the river [2]. The sediment and silt built up drying out much of Lake Conestee, which was created by the Conestee Mill to power its mill [2]. The amount of sediment and silt caused the lake to decrease in size from 145 acres to less than 18 acres [2]. While sedimentation is a concern for dams in general, the rate of sedimentation was unprecedented. Luckily, this sedimentation allowed for much of the pollutants discarded into the river earlier in the century to be buried [4]. Efforts are in place today to rebuild the Conestee Dam, that created Lake Conestee, to be able to continue to keep "2 to 3.25 million cubic yards of sediment contaminated with hazardous substates [from flowing] into the mainstem of the Reedy River" [4].

Lake Conestee and much of the surrounding areas are owned by Conestee Nature Preserve (CNP) a 640-acre nonprofit that operates the area as a wildlife sanctuary [3]. This area, with its

contamination hazards, has been transformed into habitats for many native species and wetlands that support even more species than this area used support originally. Today, the CNP has been deemed an Important Bird Area of Global Significane due to fostering a safe environment for wintering populations of a vulnerable species, the rusty blackbird [3]. Showing that good things can come from bad things in unexpected ways. Another project that is trying to help rehabilitate the river is Unity Park. A total of 8 acres of wetlands have been restored north of the park, the original path of the river before it was diverted in a 1933 project [1]. The creation of these wetlands "help in flood abatement, naturally filter harmful pollutants from water, store carbon and keep it from being released as a greenhouse gas" [1].

Weather and Water Flow

Precipitation in the Reedy River study area shows a correlation in the amount of rainfall depending on the elevation (Map 2). The higher in elevation the more rainfall the area experiences. Looking at the current precipitation rate over the past 25 years, shows variation from year to year with the range typically between 40 to 50 inches (Figure 1). The linear trend line over those past 25 years shows an increase in the amount of rainfall the area receives. While, looking further into the future shows the trend continuing with rainfall estimated to rise from an average below 48 inches, currently, to be over 50 inches in 2040 and over 50.5 inches in 2070.

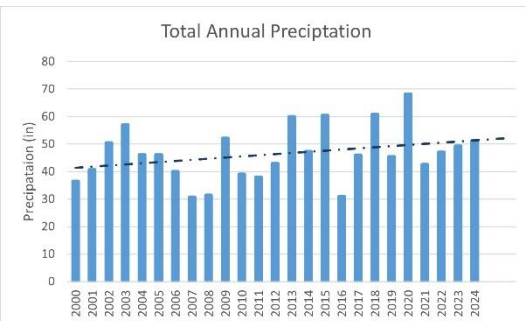
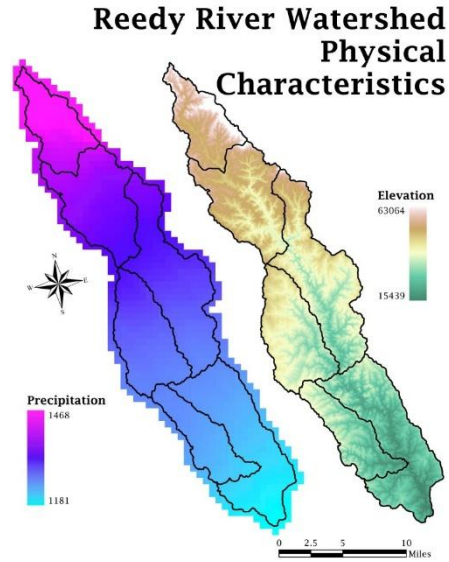


Figure 1.



Map 2.

Due to the amount of precipitation the flow rate in the Reedy River shows a similar trend (Figure 2). Selected river stream locations show that the mean flow rate will continue to increase from the historic average until 2040. The average flow rate will then decrease slightly in 2080. The best explanation for this decrease in average flow rate in 2080 can be explained through the increased variability in rainfall. While total rainfall will increase, the rain will be falling over shorter periods of time creating more intense floods. This trend of more intense rainfall events can even be seen in one day rainfall precipitation amounts (Figure 3).

Looking closer at the flow rates, the Reedy River around Greenville will see their 10-year flood rate increase from around 950 ft^3/s to 1380 950 ft^3/s in 2040. The flood rate stays the same for 2080 with the best explanation being that mitigating efforts being placed in locations like Unity Park's wetland development and other such projects further upstream of Greenville help prevent an increase by 2080. This trend is

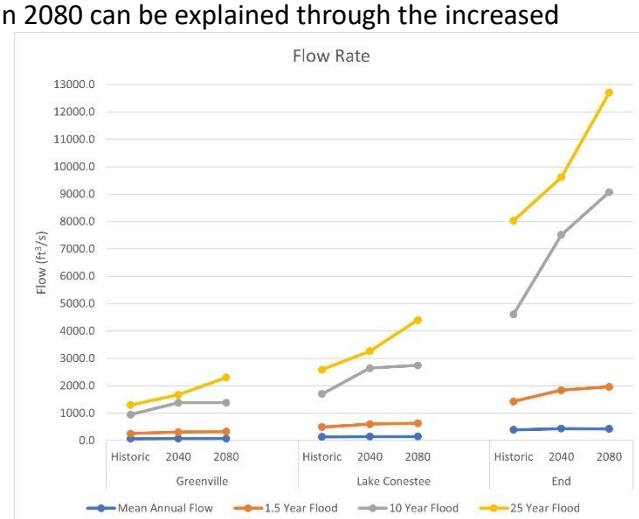


Figure 2.

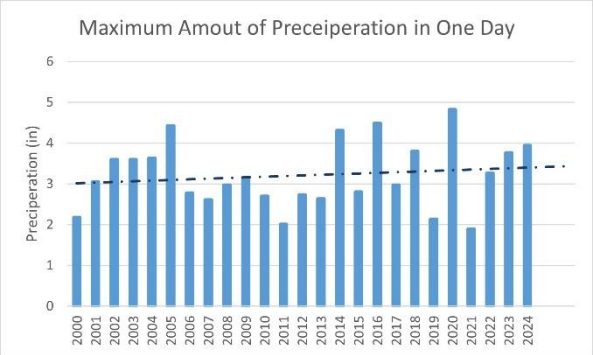


Figure 3.

similarly seen in Lake Conestee with flow rates being much higher in respect to Greenville, being that it is further downstream. At the end of the study area, it shows that while mitigation efforts to keep flood rates down from 2040 to 2080, have little impact. The most concerning fact is that the flow rate of a historic 25-year flood will become a 10-year flood in 2040 and beyond, especially in Greenville and Lake Conestee.

Rainfall of 4 inches, while only seen four times in the past 25 years, won't be that uncommon in the future. The trend line in for the amount of rainfall in one day shows an increase from about 3 inches to close to 3.5 inches. Another way to look at these heavy rainfalls is explained through looking at 2005 specifically. The amount of precipitation received in one day in 2005 of over 4 inches accounted for nearly 10% of the total amount of rainfall for the entire year. This means that just 10 days of that intensity of rain fall would be needed to get the total rainfall for the whole of 2005.

Temperature

One of the reasons that precipitation will increase, rainfall intensity will increase, and flow rate will increase all stems from an increase in temperature (Figure 4). The amount of water vaper in the atmosphere increase with temperature allowing for more intense storms and downpours. The average annual temperature from 2000 – 2023 shows a clear linear increase, with a rate of 1°F increase every 7 years. While temperature fluctuates from year to year the trend on average shows the increase. Years like 2016, 2017, and 2019 have been the warmest for this area, but if the rate of increase holds true by next year, 2026, that will be the expected average.

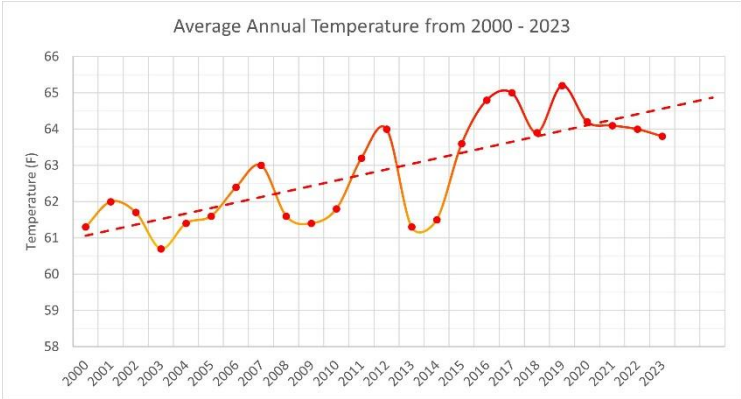


Figure 4.

Another factor that needs to be considered is that these averages come about by increasing the number of days that are spent above the current average (Figure 5). The number of days spent above 90°F on average went from about 40 in the early 2000s to around 60 in early 2020s. This is a jump of almost a week and a half dealing with temperatures over 90°F. The same trend is also seen in the number of days seen above 70°F from around 205 days in the early 2000s to 230 days in early 2020s.

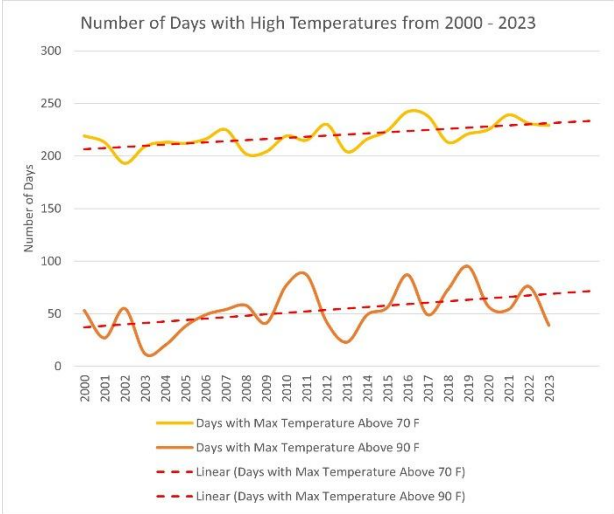
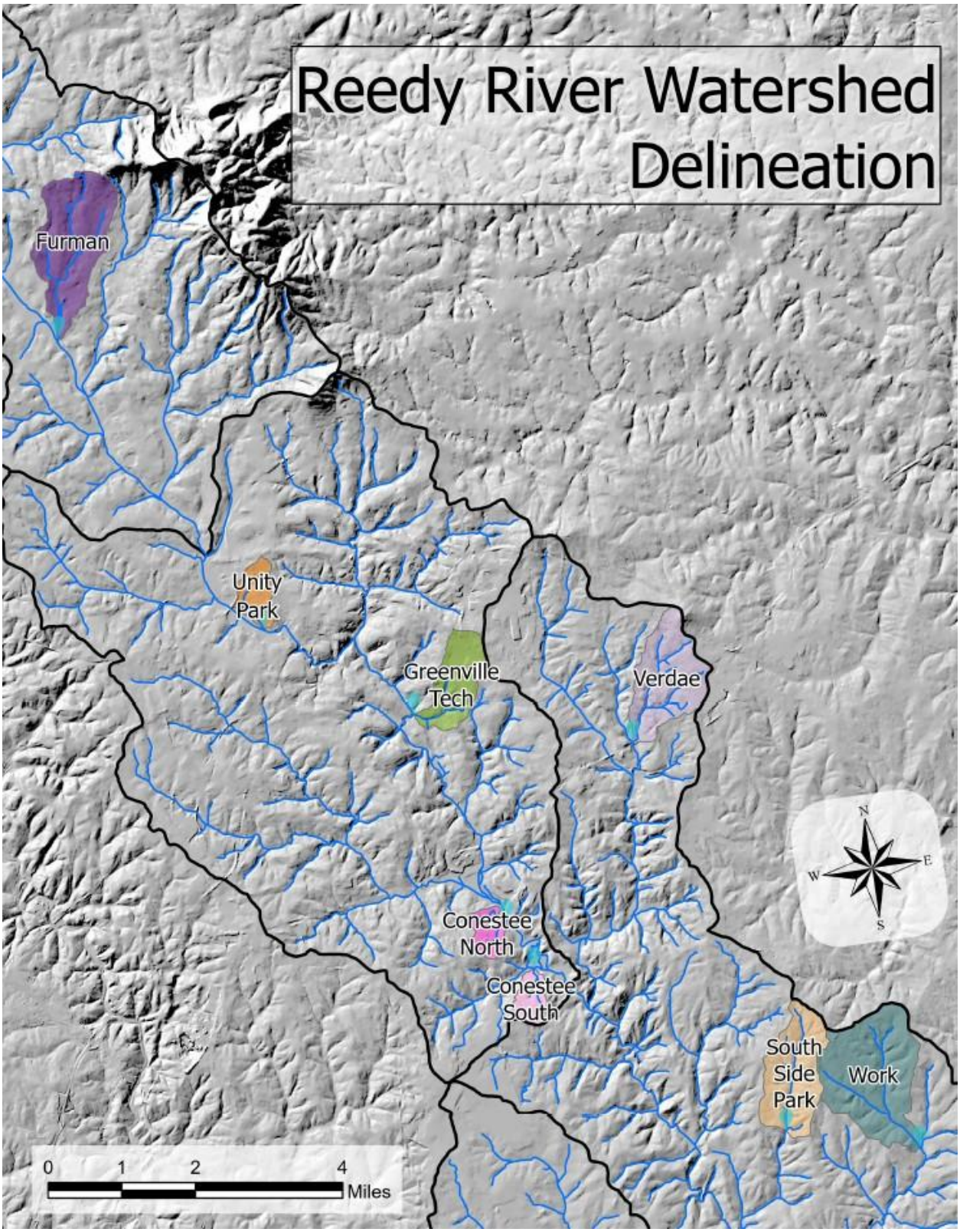


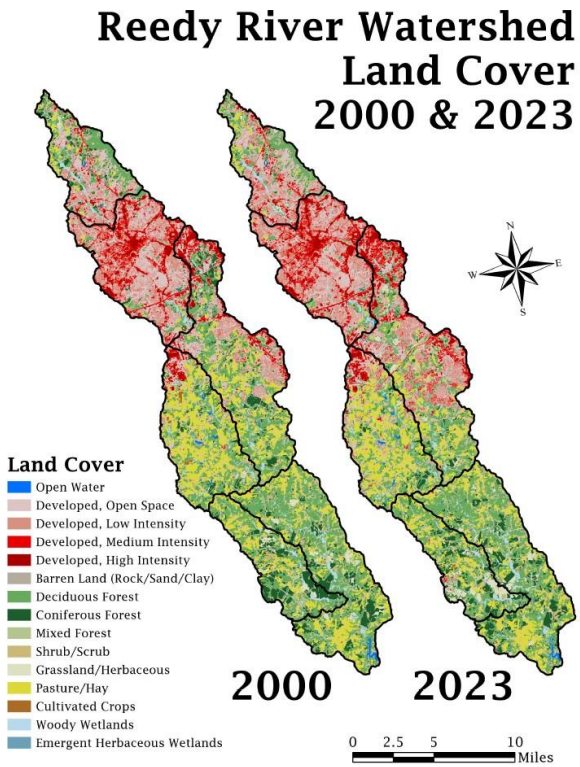
Figure 5.


Reedy River Watershed Delineation
Furman
Unity Park
Greenville Tech
Verdae
Conestee North
Conestee South
South Side Park
Work
N
E
S
W
0
1
2
4
Miles

Map 3.

Delineation

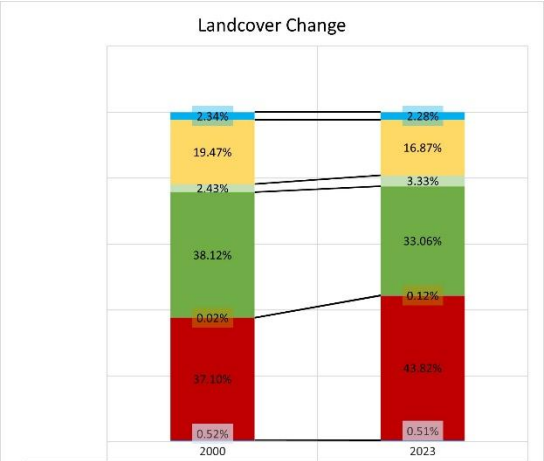
Delineation is done to be able to understand the water catchment for specific areas of the greater watershed. Map 3 shows a few areas that could be studied further to better understand how they contribute to the overall health of the Reedy River. The areas selected were those where rehabilitation efforts have been in place to see if those efforts are having a positive impact to place more of those rehabilitation efforts into other areas in the watershed. While locations were selected because of specific institutions that can be proactive in their impact when it comes to their contributions to the water flow into the river.



Map 4.

Land Cover

The land cover with in the Reedy River watershed from 2000 to 2023 had gone through a lot of urbanization (Map 4). The central area development intensity of the watershed is where the city of Greenville is located. The city between these years has only seen a slight increase in development intensity, while areas south in the suburbs have seen the most change from rural landscapes to low and medium level developments. These changes can also be understood in looking at where changes between land cover types that occurred the most frequently. This was done through reclassifying the land cover types to more general types as seen in Figure 6. The biggest transitions between land cover types between these years are forest to developed and farms to developed. This supports the real-life experience of living in the suburbs of Greenville that a lot of growth is being experienced through suburbanization. In percent of land cover types, forest went from being almost tied with developed land to now being 10% behind developed. While the land cover types like open water, wetlands, and vegetation have experienced very change.



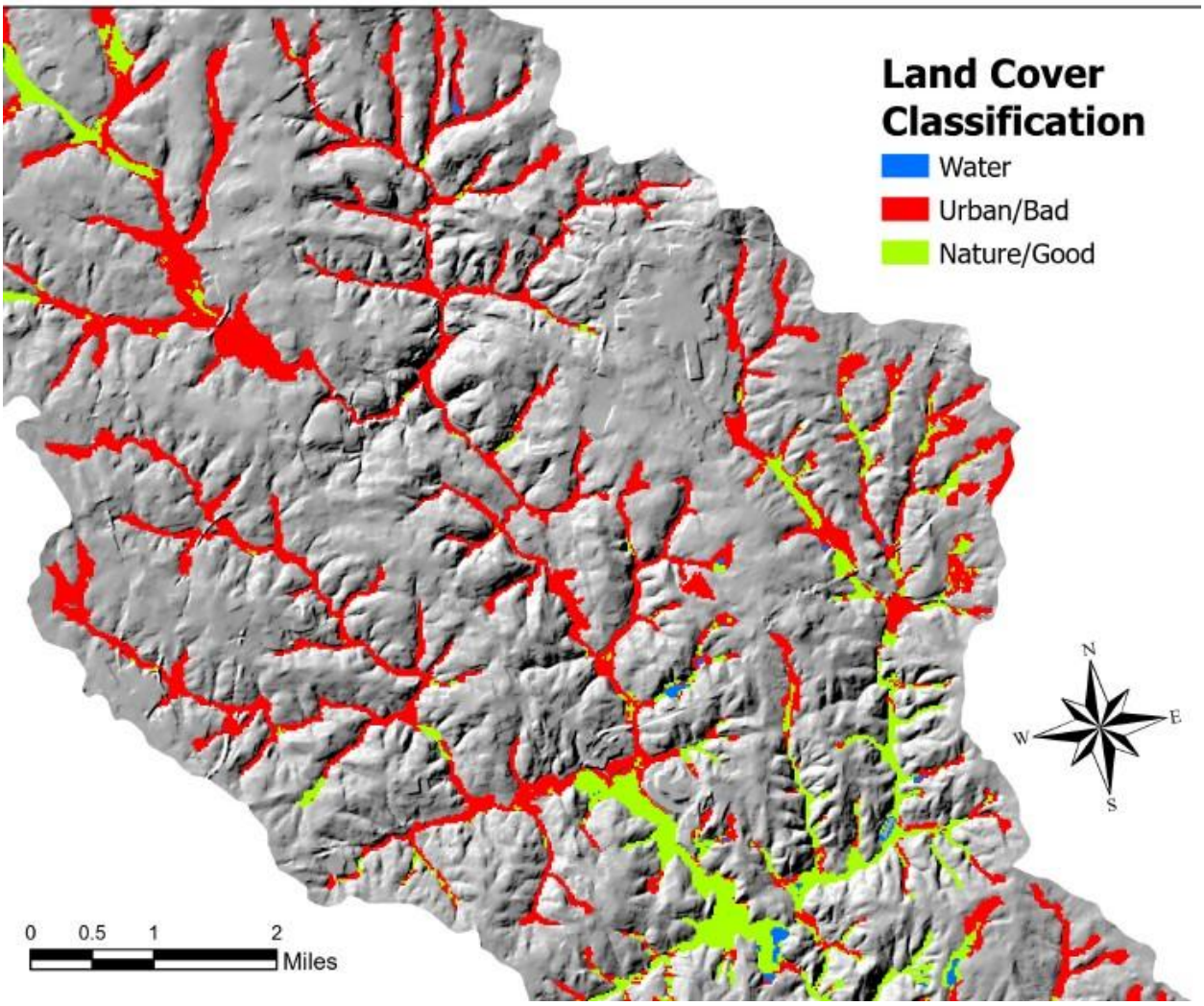
	2000	2023
Wetlands	2.34%	2.28%
Farms	19.47%	16.87%
Vegeation	2.43%	3.33%
Forest	38.12%	33.06%
Barren Land	0.02%	0.12%
Developed	37.10%	43.82%
Open Water	0.52%	0.51%

Figure 5.

Riparian

Understanding land cover types and their changes helps in understand riparian areas and their impacts. Riparian areas are those that areas around a river that impact the river the most in their contributions to runoff and impact when flooding occurs. Riparian areas are critical in helping understand the health of a river and helping improve the water quality of the river. The riparian area of the Reedy River shows the impact land cover has on the health of a river (Map 5). Most of the

Reedy River Watershed Riparian Land Cover



Map 5.

riparian area at the start of the watershed is dominated by unfavorable land cover types, developed and barren. These bad land cover types lack much of the positive that the other land cover types bring to the river. Forests and other vegetation types, for example, provide the shade and habitats for fish and other animals that rely on the river. Shade is very important, especially seen in the analysis so far, for this watershed that shows that temperatures will continue to rise. While unpleasant for humans, high temperatures can be deadly for animals that call the Reedy River home. Warmer temperatures, especially in areas lacking shade, heat up the river faster, making it harder for animals that rely on the river to continue to use it. This is why trees and vegetation are important within the riparian area. While the riparian areas downstream of the city of Greenville shows mostly good riparian land use, the expansion of suburbanization of the south of the watershed should be limited to continue to help support the river. Good riparian areas downstream of the Greenville help by providing natural filtering, especially through areas like CNP where nature is prioritized and wetlands are created.

Discussion

The Reedy River was selected for this analysis as a way to better understand the major river that is the defining feature of Greenville, South Carolina. Growing up in the area the only thing I know about the river was that it was used to power a mill or two and to not swim in it. While swimming in the river is still not recommended because of the levels of E. coli and fecal coliform that fluctuate, it has improved a lot from where it originated. The work done here is only a quick summary of the work that was done to help change the river from where it was to what it is now.

Understanding how to continue to help the river will take a lot more information and data that is available. Tests on the concentrations of different organic and non-organic materials will need to be done and most importantly understood for the native wildlife found in the Reedy River. These tests would need to be performed over a time span to confirm consistency of the results. These tests should also be performed at different sections of the river to understand how these materials are moved through the river and if there are any areas of concern that need to be looked at closer. Areas of concern can then be delineated to narrow down the location where a problem area might be located.

I can truly say that this river has improved drastically within my lifetime from being an area to avoid to an area where friends and relatives want to be around. The biggest impact has been though CNP and its wild life sanctuary. The CNP should continue its work to promote the help of the Reedy River by looking into improving land cover types along the riparian area. A collaboration could also be considered by the counties and cities that rely so heavily on the Reedy River to help preserve and prevent it from being misused, even if unintendedly from urbanization, like it was in its past.

Limitations in using some of the models for this analysis is that understanding the reasons behind different numbers aren't provided. A way to resolve this would be through the application of models are a smaller scale than those used in this analysis, which can help provide better estimates for what the future might hold in store for the Reedy River, especially when in comes to flow rates. When selecting models to showcase future scenarios I decided to use the RCP 4.5 model that is more of a middle ground scenario for climate change, rather than the RCP 8.5 that many people use as a worst-case scenario. This should be taken into account in further studies as to the use of the information. A RCP 8.5 would be beneficial when constructing near the river in case of flooding events, a RCP 4.5 would be best for understanding what everyday life will most likely be like.

Overall, the Reedy River is a beautiful river that needs to be protected. This watershed area has relied on the river, and should work on paying back to the river all the harm it did to get the city where it is now. Even small efforts like trash pick up along the shore of the river have a large impact in how the river is viewed and used by humans and animals alike.

Sources

[1] The Reedy River Wetlands. *Unity Park*. 2013. https://www.greenvillesc.gov/2013/The-Reedy-River-Wetlands

[2] River History. *Friends of the Reedy River*. https://www.friendsofthereedyriver.org/river-history

[3] Conestee Nature Preserve. *Conestee Nature Preserve.* https://www.conesteepreserve.org/

[4] About Lake Conestee Dam Restoration Project. *Lake Conestee Dam*. 2023. https://lakeconesteedam.org/about

Data Sources

Land cover and Tree Cover. https://www.mrlc.gov/data

Climate Data. https://www.ncei.noaa.gov/cdo-web/datasets.

Hydrologic Data (HUC 8 vector and HUC 4 Raster). https://apps.nationalmap.gov/downloader/.

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Precipitation models. MACA Data Portal.
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